

Guggulu and Triphala for the Treatment of Hypercholesterolaemia: A Placebo-Controlled, Double-Blind, Randomised Trial

Francesco Donato^a Elena Raffetti^b Giuseppe Toninelli^c Andrea Festa^a
Carmelo Scarcella^d Maurizio Castellano^e for the TRIGU Project Working Group

^aUnit of Hygiene, Epidemiology and Public Health, University of Brescia, Brescia, Italy; ^bDepartment of Public Health Sciences, Karolinska Institutet, Stockholm, Sweden; ^cASST Spedali Civili di Brescia, Brescia, Italy; ^dASST Del Garda, Garda, Italy; ^eDepartment of Internal Medicine, University of Brescia and Spedali Civili di Brescia, Brescia, Italy

Keywords

Clinical trial · Guggulu · Herbal medicine · Hypercholesterolaemia · Triphala

Abstract

Background: Several herbs are used for lowering high blood cholesterol levels in traditional medicines including Indian Medicine (Ayurveda). We aimed to assess the short-term effects of the combination of Guggulu (*Commiphora mukul*) and Triphala (*Terminalia chebula*, *Terminalia belerica*, and *Phyllanthus emblica*) on serum cholesterol in healthy subjects with hypercholesterolaemia. **Patients and Methods:** This was a parallel randomised double-blind controlled trial that included 90 individuals at low-moderate cardiovascular risk. The main outcome measures were serum levels of total and low- and high-density lipoprotein cholesterol (LDL-C, HDL-C). Secondary outcome measures included BMI, waist circumference, and adverse events. Subjects were administered either Guggulu and Triphala or placebo three times daily for 3 months, with 3 months of follow-up after the end of treatment. **Results:** At intention-to-treat analysis, from

baseline to 3 months, total serum cholesterol decreased by 1.9% in the placebo ($n = 44$) and 3.3% ($p = 0.01$) in the intervention ($n = 46$) group. Serum LDL-C decreased by 4.9% ($p = 0.03$) and 4.8% ($p = 0.02$) in the placebo and intervention group, respectively, without differences between them. Two participants in the intervention group developed hypersensitivity rash (4.3%) as compared with none in the placebo group. **Conclusions:** Three months of treatment with Guggulu and Triphala did not show better effects than placebo on serum levels of total and LDL cholesterol, BMI, and waist circumference.

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Guggulu und Triphala in der Behandlung von Hypercholesterinämie: Eine placebokontrollierte, doppelblinde, randomisierte Studie

Schlüsselwörter

Klinische Studie · Guggulu · Phytotherapie · Hypercholesterinämie · Triphala

Zusammenfassung

Hintergrund: In den verschiedenen Richtungen traditioneller Medizin, unter anderem in der indischen (Ayurveda), werden diverse Kräuter eingesetzt, um erhöhte Cholesterinwerte im Blut zu senken. Das Ziel unserer Studie war die Beurteilung der kurzzeitigen Wirkung der Kombination aus Guggulu (*Commiphora mukul*) und Triphala

The TRIGU Project Working Group included all the authors of the article and the following GPs: Ettore Maria Barba, Giuseppe Beltrami, Antonio De Moro, Antonia De Santis, Lucia Guarnera, Antonio Losio, Giancarlo Nervi, Erminia Pea, Rossella Pelizzari, Giovanni Rolfi, Maurizio Romano, Maria Savoldelli, Fabrizio Tanghetti, Francesca Romana Terranova, Maria Giovanna Tripaldi, Alessandro Zadra, Maria Zani.

(*Terminalia chebula*, *Terminalia belerica* und *Phyllanthus emblica*) auf den Serumcholesterinwert von gesunden Probanden mit Hypercholesterinämie. **Patienten und Methoden:** Es handelte sich um eine parallele, randomisierte, doppelblinde, kontrollierte Studie, in die 90 Personen mit niedrigem bis mittlerem kardiovaskulären Risiko aufgenommen wurden. Die wichtigsten Zielgrößen waren die Serumkonzentrationen des Gesamtcholesterins sowie des an Lipoproteine niedriger und hoher Dichte gebundenen Cholesterins (LDL-C, HDL-C). Sekundäre Zielgrößen waren BMI, Taillenumfang und unerwünschte Ereignisse. Die Teilnehmer erhielten dreimal täglich entweder Guggulu und Triphala oder Placebo für eine Dauer von 3 Monaten; nach dem Ende dieses Behandlungszeitraums folgten 3 Monate Nachbeobachtung. **Ergebnisse:** In der Intent-to-treat-Analyse für den Zeitraum von Studienbeginn bis Monat 3 ging das Gesamtcholesterin in der Placebogruppe ($n = 44$) um 1,9% zurück und in der Interventionsgruppe ($n = 46$) um 3,3% ($p = 0,01$). Das Serum-LDL-C ging in der Placebo- und der Interventionsgruppe um 4,9% ($p = 0,03$) bzw. 4,8% ($p = 0,02$) zurück; hier bestand kein Unterschied. Bei 2 Teilnehmern (4,3%) in der Interventionsgruppe trat ein Überempfindlichkeits-Ausschlag auf; in der Placebogruppe war dies bei keinem Teilnehmer der Fall. **Schlussfolgerungen:** Die dreimonatige Behandlung mit Guggulu und Triphala zeigte keine bessere Wirkung als Placebo im Hinblick auf die Serumkonzentrationen von Gesamt- und LDL-Cholesterin sowie auf BMI und Taillenumfang.

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Introduction

International guidelines report the reduction of blood total cholesterol, low-density lipoprotein cholesterol (LDL-C), or low- to high-density lipoprotein cholesterol (LDL/HDL) ratio as a milestone in the prevention of cardiovascular ischaemic disease [1]. The efficacy of statins in lowering blood cholesterol levels is well established [2, 3]. However, due to possible side effects, their use is recommended for subjects with high and very high cardiovascular risk [4, 5].

Ayurveda, the Indian traditional system of medicine, is believed to be as old as 2,500 years and it is widely practised in South Asia and increasingly in other countries, and is recognised among the “traditional medicines” by the WHO [6]. Guggulu (*Commiphora mukul*) is one of the most used herbs in Ayurveda medicine and in other traditional medicines for hypercholesterolaemia treatment. Due to its anti-hypercholesterolaemia properties [7, 8], Guggulipid was approved by the Drug Controller General of India for marketing in 1986, and now it is manufactured and marketed [9]. Both in vitro and in vivo studies de-

scribed its antagonist effect on a nuclear bile acid receptor, the farnesoid X receptor [10, 11]. Classical Ayurvedic texts recommend administration of Guggulu in combination with other herbs for modulating fatty tissue metabolism, and the Guggulu-Triphala [10] is only one of many combinations. Triphala, a complex of three plants (*Terminalia chebula*, *Terminalia belerica*, and *Phyllanthus emblica*), is used for the treatment of several clinical conditions, for its antioxidant, anti-inflammatory, and many other effects, including reduction of high serum cholesterol levels [12]. Experimental studies on animals have shown that it has cholesterol reduction properties as well as antioxidant effects [13, 14]. Since the combination of the two herbs may have more efficacy on cholesterol levels than each of them separately, a Norwegian trial investigated the efficacy of a Guggulu-, Triphala-, and other Indian plants-based formulation on blood lipids in healthy adults suggesting no change in LDL-C and a HDL-C reduction after 12 months of treatment [15].

Regarding the Guggulu efficacy in lowering high blood cholesterol, a review concluded that there is not enough scientific evidence to support its use [16]. A subsequent review argued that the cumulative data from in vitro, pre-clinical, and clinical studies largely support the therapeutic claims for Guggulu found in animal and human clinical trials conducted in India [17]. Focusing on side effects, gastrointestinal discomfort and skin rash were commonly associated with Guggulu treatment in clinical studies [15, 18]. So far, despite the lack of evidence and well-known mild and possible severe side effects, Guggulu is still used for lowering cholesterol in combination with other herbs [19]. No further clinical trial has been performed in the last decades to disentangle the possible causal effect of Guggulu treatment on lowering hypercholesterolaemia, to our knowledge. Unfortunately, a Cochrane systematic review by Gyawali et al. [20] titled “Ayurvedic herbal preparations for hypercholesterolaemia” was withdrawn in March 2019 because finishing the project within adequate deadlines could not be achieved.

The main aim of the study was to investigate the efficacy and safety of a 3-month treatment with a combination of Guggulu and Triphala on the blood levels of total cholesterol and low-density lipoprotein cholesterol (LDL-C) in subjects with hypercholesterolaemia. A secondary aim was to evaluate its effect on body mass index (BMI) and waist circumference among overweight and obese subjects.

Methods

Trial Design

A placebo-controlled, parallel-group double-blind study was performed to investigate the efficacy and safety of Guggulu and Triphala together in asymptomatic hypercholesterolaemic sub-

jects with low to moderate global cardiovascular risk (<20% cardiovascular events expected in the next 10 years). The study design was established with the support of an Ayurvedic expert physician (Dr. Deepika Gunawant).

The treatment arm included a combination of Guggulu (*Commiphora mukul*, 200 mg) and Triphala (*Terminalia chebula*, *Terminalia belerica* and *Phyllanthus emblica*, 450 mg). The parallel-group was treated with a placebo.

Active treatment and placebo capsules were provided by Dabur India Ltd., which prepared the ad hoc products for the trial. Dry resins (exudates) of Guggulu and dry powder of the Triphala pericarp were used. The extraction solvent used was methanol 100% with a herbal extract ratio of 3:1 for Guggulu, whereas water extraction (100%) was used for Triphala. Authentication of raw material and standardisation of chemical components was performed using HPLC method (details of the method are available on request). The content of Guggulu and Triphala tablets included: Guggulu extract, starch, colloidal silicon dioxide, sodium starch glycolate, sodium benzoate, potassium sorbate, microcrystalline, and Triphala extract, starch, colloidal silicon dioxide, sodium benzoate, and potassium sorbate, respectively. The quantity of active markers per dosage unit form was standardised as follows: Guggulsterone (E&Z isomer), not less than 4 mg and total tannins, not less than 40 mg for Guggulu and Triphala, respectively. The content of placebo tablets was: calcium carbonate, starch, talc, magnesium stearate, microcrystalline cellulose, and colloidal silicon dioxide. Both Guggulu and Triphala and placebo tablets were analysed for the presence of heavy metals and microbiological contamination and were certified to be free.

Participants

The eligibility criteria of patients are set out in Table 1. A total of 17 GPs participated in the project. Informed written consent was provided by each patient. The study was performed in agreement with the Helsinki Declaration for clinical trials in human beings. Patients' data were treated in agreement with the Italian laws on privacy protection. The project was approved by the Ethics Committee of the Local Health Authority (LHA) of Brescia, Italy, on 21 June 2007.

The study took place in the 17 GPs' practices in Brescia province, Northern Italy, from June 2009 to November 2010.

Interventions

After an initial screening visit, participants were randomised in a 1:1 fashion to receive one capsule of Guggulu, 200 mg, and one of Triphala, 450 mg, or placebo three times daily, immediately after breakfast and principal meals for 3 months. The follow-up was 3 months after the end of the treatment. A computer-generated random number list was prepared by a data analyst in the coordinating centre (director: F.D.) with no clinical involvement in the trial. To minimise group imbalances, randomisation was also blocked using equal blocks of 4, using a permuted block design with a computer random number generator. The allocation list was locked away in the coordinating centre and was inaccessible to GPs, patients, and laboratory staff.

All the GPs and patients were blinded to treatment assignment and block size (double-blind). Herbal products and placebo tablets were identical in appearance (no difference in size, colour, and smell). They were pre-packed in identical bottles, which were consecutively numbered according to the randomisation schedule. Each patient was assigned a code on the basis of the randomisation list and received the capsules in the corresponding bottle. All participants were told to maintain their usual dietary habits.

Study assessments took place at baseline, at 1 month after initiation of treatment, at 3 months (end of treatment), and at 6

Table 1. Eligibility criteria

Inclusion criteria	
Signed written informed consent	
Global cardiovascular risk: <20% of events expected in the next 10 years	
BMI ≥ 18.5	
Total serum cholesterol 200–350 mg/dL	
Therapy-free interval of 6 months or longer	
Aged 35–69 years	
Ability to speak, read, and write Italian	
Willingness to prevent conception during the study	
Exclusion criteria	
Documented hypersensitivity to the study treatment or its excipients	
History of ischaemic disease	
History of diabetes mellitus	
Global cardiovascular risk: $\geq 20\%$ of events expected in the next 10 years and/or statin treatment	
Presence of a severe chronic disease	
Chronic therapy with any drug	
Pregnancy or breastfeeding	

months (3 months after the end of the treatment). At each study visit, participants were weighed on a standardised scale, had full vital signs measured, and were asked about adverse events and possible changes in medications. The capsules possibly not assumed were counted and the reason for not taking the capsules was asked. A few days before the visit, subjects gave blood samples for lipid and chemistry analyses so as the results of the analysis were available and discussed with the GP during the visit. At baseline and end-of-treatment visits, patients also provided an overnight urine sample. For each patient, an evaluation of his or her global cardiovascular risk was performed using the Italian Health Ministry computer programme or cards [21] according to inclusion criteria. BMI was computed as the ratio of weight (in kg) on squared height (in metres). Waist circumference was measured in the standing position, midway between the lowest rib and the iliac crest, using a standard tape [22] according to recommended methods of adult anthropometry for public health and primary health care [23].

All the laboratory tests were performed at the Laboratory of the Brescia LHA. Lipid parameters were analysed from EDTA plasma collected after a 12-hour fast. Plasma levels of total cholesterol, high-density lipoprotein cholesterol (HDL-C), and triglycerides were measured enzymatically on a Cobas INTEGRA 400 Plus (Roche Diagnostic Systems Inc., Indianapolis, IN, USA) using Roche reagents.

Levels of LDL-C were determined using the formula: total cholesterol – HDL-C – $1/5 \times$ triglycerides. Furthermore, sera were assayed for glucose, alanine aminotransferase (ALT), lactate dehydrogenase (LDH), creatine phosphokinase (CPK), and creatinine concentration on a Cobas INTEGRA 400 Plus (Roche Diagnostic Systems Inc., Indianapolis, IN, USA) using Roche reagents. A whole blood sample was also collected and red and white blood cell counts and formula were determined using Beckman Coulter HMX. A complete urine analysis was also performed with Aution sticks use by visual interpretation.

For each reported adverse event, study investigators judged and recorded the severity, relationship to study agent, and action taken. On the basis of type and severity of the event a decision was taken by the GP together with the principal investigator as regards the possible interruption of treatment and prescription of medicines.

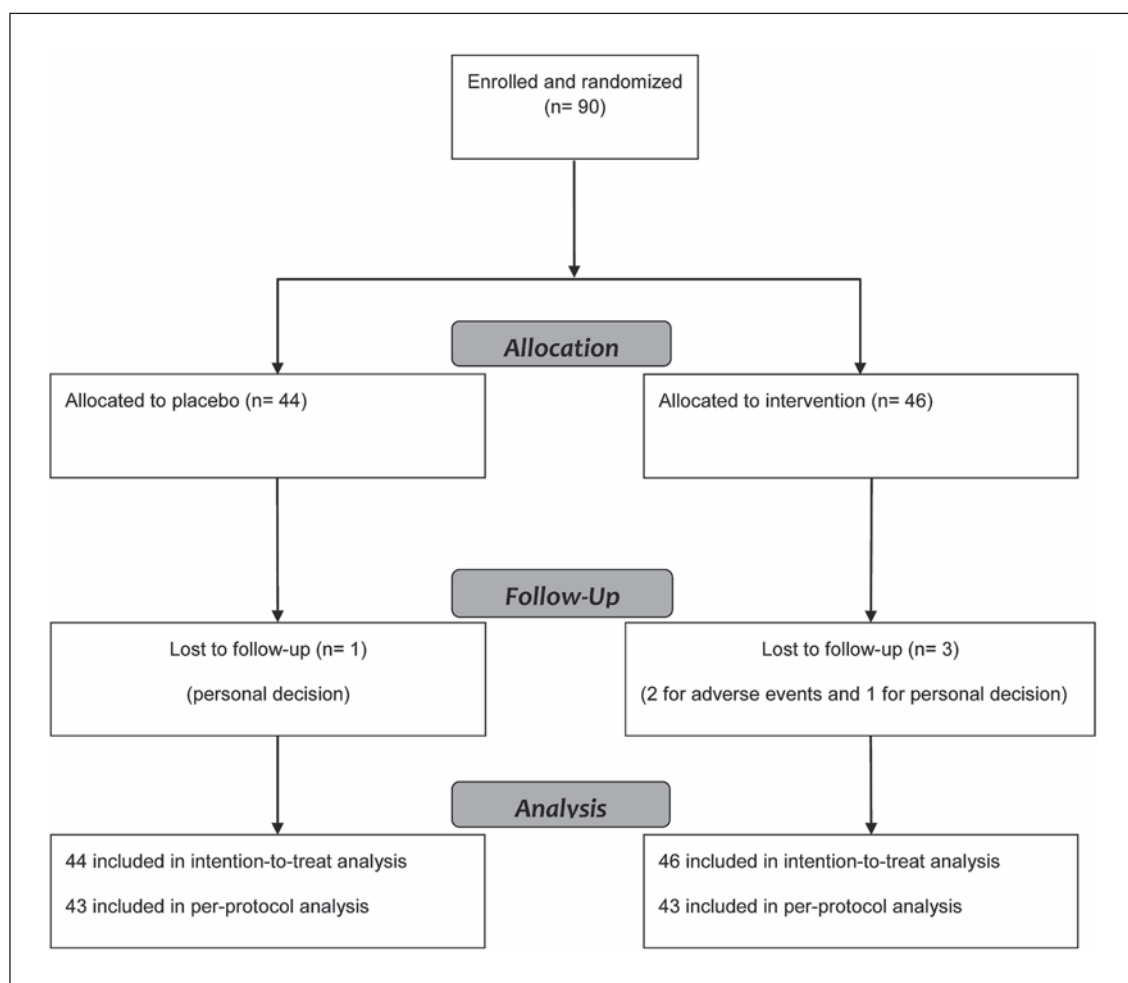


Fig. 1. Study flow diagram.

During the last visit, patients filled in a short questionnaire on their perceived health status, satisfaction level on having participated the trial, and their belief whether they had received the active product (herbs) or placebo.

Outcomes

The primary endpoint concerning the efficacy of the treatment was a reduction in serum levels of total cholesterol and LDL-C. Secondary endpoints included measures of BMI, waist circumference, and adverse event reports. The following safety biological parameters were also measured: serum levels of glucose, triglycerides, LDH, CPK, ALT, creatinine, standard whole blood examination (red and white cells, haemoglobin, haematocrit, etc.), and standard analysis of overnight urine sample (pH, specific gravity, glucose, proteins, haemoglobin, ketone, bilirubin, urobilinogen, nitrites, and microscopic analysis of urine sediment).

Sample Size

The study was designed to observe a significant decrease in blood total cholesterol of at least 10% in the treatment versus no change in the placebo group at the end of the trial, assuming a total blood cholesterol mean of 250 mg/dL in enrolled subjects, with a standard deviation (SD) of 25. Accounting for 10% dropouts, we estimated that a sample size of 45 per group would provide at least 90% power to detect 10% decrease in total cholesterol in the intervention group and 10% difference between the 2 groups at the end

of the treatment, using a 2-tailed test with an alpha value of 0.05 and an estimated within-group SD of 10%.

Statistical Analysis

The primary analysis was by intention-to-treat, using the last observation carried forward for all missing lipid parameters. For continuous variables, the differences between basal and after-treatment values in both groups (intra-group analysis) and between the intervention and control group at baseline and end of treatment (inter-group analysis) were performed using *t* tests for paired and unpaired data. Two-way analysis of variance (ANOVA) was also performed for testing both time (baseline vs. end of treatment) and group (intervention vs. placebo). All the analyses concerning lipid and other serum parameters and whole blood and urine variables were performed on log-transformed values whenever they showed a better approximation to a normal distribution than original figures. Differences in lipid variables, BMI, and waist circumference were also analysed in the subgroup of overweight and obese subjects (BMI ≥ 25). For ease of interpretation, changes of the raw data are presented for these log-transformed variables. We also analysed the proportion of subjects who experienced the beneficial effect of treatment, defined as a decrease of $>10\%$ in serum total and LDL cholesterol, of ≥ 1 unit (kg/m²) of BMI and ≥ 1 cm in waist circumference. BMI was categorised to classify subjects as normal (<25), overweight (≥ 25 and <30), and obese (≥ 30).

Table 2. Baseline characteristics of study participants

Parameter	Placebo (<i>n</i> = 44)	Intervention (<i>n</i> = 46)	All subjects (<i>n</i> = 90)
Age, years			
Mean (SD)	54 (8.6)	52.19 (9.5)	53.17 (9.1)
Range	36–69	28–68	28–69
Gender			
Males, <i>n</i> (%)	13 (29.9)	14 (30.4)	27 (30.0)
Females, <i>n</i> (%)	31 (70.1)	32 (69.6)	63 (70.0)
Total cholesterol, mg/dL			
Mean (SD)	264.7 (36.0)	267.35 (36.8)	266 (36.3)
Range	201–350	203–340	201–350
LDL-C, mg/dL			
Mean (SD)	179.0 (32.9)	182.43 (37.2)	180.8 (35.2)
Range	121.2–278.8	124.2–271.2	121.2–278.8
BMI			
Mean (SD)	27.4 (3.3)	27.85 (4.1)	27.6 (3.7)
Range	19.6–35.2	18.9–30.2	18.9–35.2
BMI <25, <i>n</i> (%)	8 (18.2)	12 (26.1)	20 (22.2)
BMI ≥25, <i>n</i> (%)	37 (81.8)	34 (73.9)	71 (78.9)
Waist circumference, cm			
Mean (SD)	95.1 (10.9)	95.39 (12.2)	95.26 (11.5)
Range	65–115	64–126	64–126
Males >102 cm, <i>n</i> (%)	7 (53.9)	6 (42.9)	13 (48.2)
Females >88 cm, <i>n</i> (%)	22 (71.0)	22 (68.7)	44 (69.8)

LDL-C, low-density lipoprotein cholesterol; BMI, body mass index.

All the analyses were performed using the Stata programme (Stata Statistical Software release 10.0, 2009; Stata Corporation, College Station, TX, USA).

All the statistical tests are 2-tailed with *p* value of 0.05 for refusing the null hypothesis.

Results

Patient Characteristics

The flow chart of the study participants is illustrated in Figure 1. Over a 12-month period, 90 healthy individuals, none of which had cardiovascular or other chronic diseases, accepted to participate among those considered eligible on the basis of inclusion and exclusion criteria. Eighty-six of 90 participants completed all study-related visits (4% dropout rate): 43 of 44 (98%) participants assigned to receive placebo and 43 of 46 (93%) participants assigned to the intervention group completed the study (*p* > 0.1).

Subjects' characteristics and baseline values of the enrolled participants are shown in Table 2. Among them, 70% were women and the mean (SD) age was 53.2 (9.1) years. Total and LDL cholesterol means (SD) were 266 (36.3) and 180.8 (35.0) mg/dL. The mean (SD) BMI was 27.6 (3.7), with 78.9% of subjects being overweight or obese (BMI >25), and the mean (SD) of waist circumference was 95.3 (11.5) cm, with 48.2% males and 69.8% fe-

males being over the suggested cut-offs of 102 and 88, respectively, for Western populations. There were no statistically significant differences in any of the demographic, physical, and biological parameters between the two groups at baseline.

Effects on Total and LDL Cholesterol Levels and Physical Parameters

According to intention-to-treat analysis (*n* = 90), from baseline to 3 months of treatment, the levels of total and LDL cholesterol decreased, respectively, by 1.9% and 4.9% (*p* = 0.026) in the placebo group, and 3.3% (*p* = 0.013) and 4.8% (*p* = 0.017) in the intervention group (Table 3; Fig. 2). In the same period, serum concentration of HDL increased significantly in the placebo (+8%) but not significantly in the intervention group (+1.4%). No significant changes were observed in triglycerides in both groups, whereas a significant decrease was observed in BMI (−1.1%, *p* = 0.05) and waist circumference (−1.6%, *p* = 0.02) in the intervention but not placebo group.

In the per-protocol analysis (*n* = 86), total and LDL cholesterol levels decreased by 1.9% and 4.7% (*p* = 0.03) in the placebo and 3.5% (*p* = 0.013) and 5.3% (*p* = 0.013) in the intervention group, respectively; BMI did not change, whereas waist circumference was reduced significantly (−1.5%, *p* = 0.03) in the intervention group only (data not shown in table).

Table 3. Results of intention-to-treat analysis ($n = 90$) for lipoprotein parameters, BMI, and waist circumference at the end of the treatment (3 months)

Parameter	Placebo (<i>n</i> = 44)			Intervention (<i>n</i> = 46)			Means difference (placebo – intervention) (95% CI)	<i>p</i> value		
	Baseline, mean (SD)	Month 3, mean (SD)	Difference (%)	<i>p</i> value	Baseline, mean (SD)	Month 3, mean (SD)			Difference (%)	<i>p</i> value
Total cholesterol, mg/dL	264.7 (36.0)	259.6 (35.9)	–5.0 (–1.9)	NS	267.3 (36.8)	258.5 (38.8)	–8.9 (–3.3)	0.013	1.16 (–14.52, 16.84)	NS
HDL-C, mg/dL	58.4 (13.4)	63.1 (18.5)	+4.7 (+8.0)	0.008	55.8 (12.0)	56.6 (12.3)	+0.8 (+1.4)	NS	6.50 (–0.04, 13.04)	NS
LDL-C, mg/dL	179.0 (32.9)	170.3 (33.1)	–8.7 (–4.9)	0.026	182.4 (37.2)	173.7 (35.4)	–8.8 (–4.8)	0.017	–3.37 (–17.74, 10.99)	NS
Triglycerides, mg/dL	136.0 (84.1)	132.3 (49.3)	–4.7 (–3.4)	NS	145.7 (88.4)	141.2 (78.4)	–4.5 (–3.1)	NS	–9.85 (–37.44, 17.73)	NS
BMI	27.4 (3.3)	27.3 (3.4)	–0.1 (–0.3)	NS	27.8 (4.1)	27.5 (4.0)	–0.3 (–1.1)	0.059	–0.22 (–1.78, 1.33)	NS
Waist circumference, cm	95.1 (10.9)	95.0 (10.5)	–0.1 (–0.1)	NS	95.4 (12.2)	93.9 (11.9)	–1.5 (–1.6)	0.020	1.10 (–3.66, 5.86)	NS
HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; BMI, body mass index; NS, not significant, <i>p</i> > 0.1.										

No statistically significant differences in total and LDL cholesterol levels, BMI, and waist circumference were found between the intervention and placebo groups after 3 months of treatment in both intention-to-treat and per-protocol analyses.

According to the definition of a favourable response to treatment (see Methods), the intention-to-treat analysis showed positive effects in total cholesterol, LDL-C, BMI, and waist circumference in 18%, 32%, 14%, and 45% in the placebo, and 28%, 37%, 13%, and 59% in the intervention group, respectively ($p > 0.1$ for each comparison between the two groups).

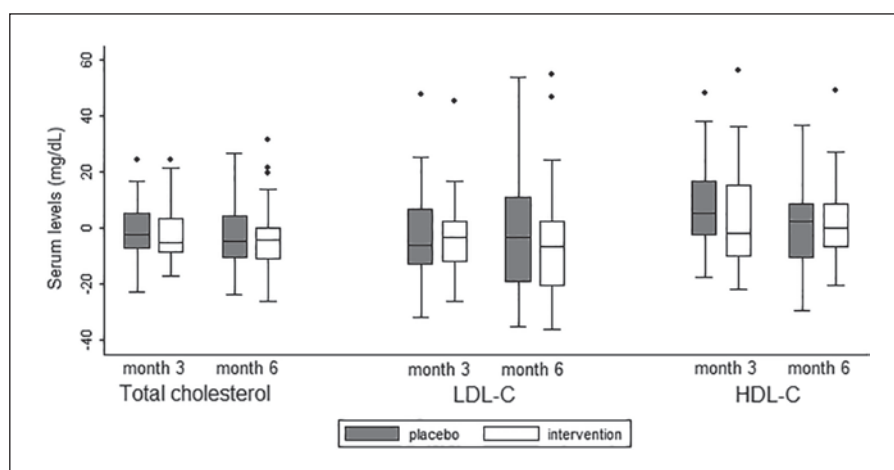
At the 6-month visit, corresponding to the end of follow-up (3 months after the end of treatment), according to intention-to-treat analysis, the levels of total and LDL cholesterol decreased, with respect to baseline values, by 3.4% ($p = 0.05$) and 4.4% ($p = 0.02$) in the placebo group, and by 5.4% ($p = 0.005$) and 8.2% ($p = 0.01$) in the intervention group, respectively (data not shown in table). Serum concentrations of HDL and triglycerides, and BMI and waist circumference did not vary significantly with respect to baseline, in each group. No statistically significant differences were found in changes of total and LDL cholesterol levels between baseline and 6 months between the two groups. Similar results were observed in the per-protocol analysis and in the analysis restricted to overweight and obese subjects.

The analysis of participants with favourable response to treatment at 6 months compared to baseline showed positive effects in total cholesterol, LDL-C, BMI, and waist circumference in 30%, 45%, 16%, and 41% in the placebo and 37%, 52%, 23%, and 54% in the intervention group, respectively ($p > 0.1$ for each comparison between the two groups).

Effects on Other Biological and Physical Parameters

There were no significant changes overall and in each group in serum levels of glucose, LDH, CPK, red and white blood cell counts, and urine parameters from baseline to 3 months. On the contrary, statistically significant decreases were observed in serum creatinine and ALT in both groups at intention-to-treat analysis: creatinine decreased by 4.9% and 5.1%, and ALT by 20.2% and 30.4%, in the placebo and intervention group, respectively. No statistically significant differences were found in serum creatinine and ALT values at 3 months between the two groups. Similar patterns were observed in overweight and obese subjects: creatinine decreased significantly by 6% and 7.2%, and ALT by 16.5% and 29.8% in the placebo and intervention group, respectively, with no statistically significant differences between the two groups at 3 months.

Fig. 2. Box-and-whisker graphs of percentage changes from baseline to 3 months (end of treatment) and 6 months (end of follow-up) in lipoprotein parameters, in the placebo and intervention groups, at intention-to-treat analysis. HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol.



Safety and Tolerability

There were no changes in renal function or electrolytes in both groups after 3 months. However, hypersensitivity rash was observed in 2 participants in the intervention group (4.3%), which occurred some minutes after capsule ingestion, with facial oedema in one of them, and was judged as possibly related to herbal treatment. The patients stopped treatment and were treated with oral steroids for 2 days, and the rash disappeared. They were considered dropouts from the study.

A few patients reported temporary loose stools or moderate diarrhoea in both groups. The symptoms were moderate and well tolerated; they disappeared spontaneously in a few days and did not require the withdrawal of treatment.

Adherence and Masking

At the end of the study, the participants who completed treatment and follow-up ($n = 86$) filled in a short questionnaire concerning their beliefs about the study. 89% of them said to have regularly taken therapy, 81% did not report any change in their health status during or after the study, 92% reported no health problems with treatment. 72% of patients said they did not know whether they had received the herbal products or placebo, and 15% guessed treatment and 13% placebo group, without differences between the two groups (12% and 18% thought they had received herbal treatment in the placebo and intervention groups, respectively, $p > 0.1$), suggesting adequate masking of the study agent.

Discussion

This trial complies with the preferred reporting items of CONSORT 2010 Statement [24] and CONSORT Statement of Herbal Medicines [25].

In this randomised clinical trial, a combination of Guggulu and Triphala did not improve total and LDL cholesterol serum levels better than placebo in asymptomatic hypercholesterolaemic subjects at low to moderate cardiovascular risk, who followed a Western diet. Both intervention and placebo groups showed a slightly improved lipidemic pattern after 3 months of treatment. The decreases in total and LDL cholesterol observed in the intervention group of -3.3 and -4.8% , respectively, were lower than the 10% expected as the main objective of the trial. This cannot be due to low statistical power, as our sample size was large enough to detect a 10% reduction of total or preferably LDL cholesterol, should it have occurred. Similar changes were observed in the placebo group, which were possibly due to patients' lifestyle changes, especially in dietary habits and physical exercise, although subjects had been told not to change their habits during the trial. We also observed a modest increase of HDL-C in the control but not in the intervention group. This difference may probably be considered an effect of Guggulu, which reduced serum levels of HDL-C in other trials [15, 18].

The main findings of our study are in agreement with two other randomised double-blind trials conducted in Western countries [15, 18], though the USA study used an extract from Guggulu (guggulsterones) [18] and the Norwegian one used Guggulu together with other herbs, including Triphala [15]. They are in contrast with previous trials performed in India, most of which showed 10–30% reduction in total cholesterol and triglycerides, with 60–80% participants showing a favourable response rate in target parameters [26–29]. Actually, most of Indian trials were of poor quality as they did not include a control group, were not double-blind or randomised, or they were small [16] given that an association between trial dimension and methodological quality has been reported [30]. However, the discrepancies between the results of

Indian and Western studies may at least partly depend on the differences between Indian and Western populations as regards dietary, lifestyle, and genetic characteristics that may affect lipid metabolism and biological response to herbal medicine active constituents. Traditionally, Indian people were thinner and physically more active and ate less total fat, less animal food, and more dietary fibre than Western populations. In fact, one study showed beneficial effects of guggul in non-obese but not in obese subjects [29]. Also, different formulations, dosages, and active principles of guggul used in the different studies may explain the different findings.

These negative results are also in contrast with findings from experimental studies, which showed a consistently hypolipidemic effect of guggulipid and guggulsterones in various animal species, including rat, mouse, rabbit, chicken, domestic pig, dog, and monkey [17]. Furthermore, *in vitro* and *in vivo* studies identified various potential mechanisms of action of guggulsterones. In particular, the bioactive constituent of Guggulu is an antagonist at the nuclear farnesoid X receptor, a key transcriptional regulator for the maintenance of cholesterol and bile acid homeostasis [10, 11]. This is a potent antagonist at the mineralocorticoid, glucocorticoid, androgen, progesterone, and oestrogen receptors. It also up-regulates the expression of the bile salt export pump, a rate-limiting efflux transporter for eliminating cholesterol metabolites with bile acids from the liver. One possible explanation suggested by Szapary et al. [18] for the contrasting results between experimental studies and clinical trials in humans is that guggulsterones may have both antagonist and agonist activity on FXR, and/or that they, like phytoestrogens, may have different lipid effects in different species and populations.

Possible adverse effects and pharmacological interactions of treatments with herbal products remain of concern [31].

A few patients reported minor gastrointestinal symptoms, such as loose stools, diarrhoea, and/or abdominal discomfort, which have been reported in other studies [15, 18, 27, 28] and did not require interruption of the treatment. The most important side effect observed in our study was skin rash, which occurred in two subjects (4.3%), with facial oedema in one of them, a few minutes after the capsule ingestion. It was treated with oral corticosteroids and led to the interruption of the treatment. According to other reports [15, 18, 32], it seemed to be a hypersensitivity drug reaction probably caused by Guggulu, whereas the use of Triphala has not been associated with hypersensitivity reaction so far, to our knowledge. No side effects have been observed on renal and liver functions, haematological parameters, and urine electrolytes in agreement with previous studies, [18, 26], though one case of rhabdomyolysis possibly associated with Guggulu was reported.

Some potential limitations of our study should be taken into account. First, 3 months of treatment may not be a long enough time to see the benefit of this treatment. However, the duration of treatment in the trials that showed beneficial effects of Guggulu on total cholesterol and LDL-C [26–29] was no longer than this research. Second, we did not suggest our patients any specific diet. Our goal was to evaluate the pure effect of Guggulu and Triphala combination and therefore we asked participants to maintain stable dietary habits, which they confirmed to their GPs at the end of the study. Accordingly, most of the previous studies showing the benefit of Guggulu did not include dietary recommendations [27]. Third, the daily dose of Guggulu and Triphala in our trial may not have been high enough to produce a cholesterol reduction. We administered the dosages of Guggulu and Triphala included in commonly available dietary supplements in Western countries, with a content of active principles such as guggulsterone and tannins similar to that used in other clinical trials. However, a higher dosage may be required for effective treatment of hypercholesterolaemia, although the risk of side effects may be higher too. Finally, the true effects of the herbal treatment alone may be lower than those we had hypothesised, and therefore a greater number of subjects should have to be enrolled to detect lower differences between the groups. However, less than 10% reduction in total and LDL cholesterol may be a too modest clinical target to deserve another trial. Anyway, this study did not aim to test the efficacy of Ayurvedic medicine in the treatment of hypercholesterolaemia, since it is a complex system of medicine that never recommends the use of herbal products alone but suggests taking them together with physical treatments, dietary recommendations, and physical activity [33].

The Triphala-Guggulu combinations are used in traditional Ayurvedic medicine, as described in *sharangdharsanhita*, and recently prepared in standardised, stable guggulkalpa tablets [10]. However, the traditional Ayurvedic approach bases its diagnosis and therapy on Ayurvedic paradigms and not on conventional diagnoses. The Ayurvedic treatment is strongly individualised as it varies according to patient's constitution ("prakriti") and his/her specific equilibrium alterations ("vikriti"), so as no "standard treatment" for hypercholesterolemia really exists in Ayurveda medicine [33].

In conclusion, this study does not support the use of Guggulu and Triphala-based formulations for the treatment of hypercholesterolaemia and overweight or obesity in a Western population, although a supportive role of these herbs in a more comprehensive approach, including also lifestyle changes and physical treatments, as it is in the usual practice in Ayurveda medicine, cannot be excluded.

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Statement of Ethics

The study was conducted in accordance with the guidelines of the Declaration of Helsinki and the principles of Good Clinical Practice. Written informed consent was obtained from all subjects. The study protocol was approved by the local Ethic Committee. The trial was registered at Agenzia Italiana del Farmaco (Italian Medicine Agency), Eudract No. 2008-008376-14.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

F.D., as principal investigator of this study, had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analyses. Study concept and design: F.D. and M.C. Acquisition of data: C.S. and G.T. Analysis and interpretation of data: A.F., F.D., M.C., E.R. Drafting of the manuscript: F.D., M.C., E.R. Critical revision of the manuscript for important intellectual content: F.D., M.C., E.R. Statistical analysis: A.F. Obtained funding: F.D. Administrative, technical, and material support: C.S. and G.T. Final approval of the version to be published: all authors. The GPs who adhered the TRIGU Project Working Group selected and recruited patients, administered products, required laboratory examinations, and followed the patients according to the study protocol.

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